

```
1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity reg is
5      port(rin: in std_logic_vector(3 downto 0);
6           en, clk: in std_logic;
7           rout: out std_logic_vector(3 downto 0));
8  end entity;
9
10 architecture r of reg is
11     signal rt: std_logic_vector(3 downto 0);
12 begin
13     process(clk)
14     begin
15         if(falling_edge(clk)) then
16             if (en = '1') then
17                 rt <= rin;
18             end if;
19         end if;
20     end process;
21     rout <= rt;
22 end r;
```